Finite-Horizon Hankel Profiles

Separate Temporal Depth from Effective Hankel Dimension

Standard spectral summaries can match even when recurrent systems remember differently.



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CENTRAL IDEA • Recurrent memory has two coordinates: WHEN influence appears (Dε) and HOW MANY modes carry it (rτ(T)).

01 Motivation & research question

Recurrent systems store past inputs and use them later but memory has two distinct aspects. A complete description must tell us both when a past input becomes visible and how concentrated that influence is.

WHEN?

Temporal depth Dε

How late does an input affect the output?

HOW MANY?

Effective dimension rτ(T)

How many dominant patterns carry its energy?

An echo analogy



Like an echo: two rooms can reflect the same total sound energy, yet one returns the echo quickly while the other delays it.
Total energy alone does not tell us when the response arrives.

Research question

Can ordinary summaries—eigenvalues, singular values, or total / lag-wise energy—fully describe recurrent memory?
Or can two systems look identical under those summaries while still differing in when information appears and how many dominant response patterns carry it?

Our answer: No.

Two explicit analytic constructions show that the answer is no.
One keeps familiar spectral summaries fixed but changes memory timing; the other keeps lag-wise energy fixed but changes effective Hankel dimension.

02 Two precise memory coordinates

Linear recurrent core (one input, one output)

$$h_{t+1} = A\,h_t + B\,x_t \qquad y_t = C\,h_t$$

Response after k steps: $K_k = C\,A^k\,B$
These Markov parameters are the input–output sequence used throughout the poster.

Organize these responses into a Hankel matrix

K0	K1	K2
K1	K2	K3
K2	K3	K4

$$H_t = [\,K_{i+j}\,]_{i,j=0}^{T-1}$$

Its singular values $\sigma_1 \geq \sigma_2 \geq \ldots$ summarize how strongly the finite-horizon Hankel energy is distributed across leading modes.

Energy at a finite horizon vs. total energy

$$E_t = \|H_t\|_F^2 = \Sigma_i\,\sigma_i^2$$

$$E^\infty = \Sigma_{k\geq 0}\,(k+1)|K_k|^2$$

Temporal depth — WHEN

$$D\varepsilon = \min\{\,T : E_t \geq (1-\varepsilon)\,E^\infty\,\}$$

First horizon capturing at least $1-\varepsilon$ of total energy.

Effective Hankel dimension — HOW MANY

$$r\tau(T) = \min\{\,r : \Sigma_{i=1}^{r}\,\sigma_i^2 \geq \tau\,E_t\,\}$$

Fewest leading modes needed to capture at least a τ fraction of the finite-horizon energy by horizon T.

03 Result 1 · Same spectra, different timing

For every $n \geq 3$, two stable systems can have identical B, C, total Hankel energy, and matching spectral properties of A yet their dependency horizons differ by Ω(n). So standard spectral agreement does not determine when memory becomes visible.

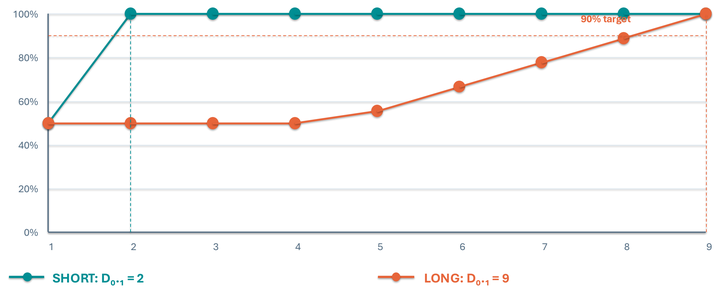
SHORT
Extra response at lag 1



LONG
Extra response at lag n − 1



Fraction of total energy E_t / E^∞ (n=9, δ=1, ε=0.1)



Both systems share an immediate response; only the extra response arrives later.
The delayed lag cannot enter H_t until $2T - 2 \geq n - 1$, so the long system hides its extra response until larger horizons.
This is why matching A-spectra and total energy still does not determine WHEN memory appears.

04 Result 2 · Same lag energy, different mode counts

A sign change can preserve $|K_k|^2$ at every lag while changing how the energy is distributed across Hankel modes.
Thus equal lag-wise energy does not force the same effective dimension.

PLUS (+)

$K^+ = (1, 1, +1, 0, \ldots)$

$H =$

1	1
1	1

$\sigma = (2, 0)$

MINUS (−)

$K^- = (1, 1, -1, 0, \ldots)$

$H =$

1	1
1	-1

$\sigma = (\sqrt{2}, \sqrt{2})$

Mode-energy share (T=2)

PLUS · 100% in one mode



MINUS · 50% / 50%



At τ = 0.8 and T = 2: $r_{0.8}^{+}(2) = 1$ versus $r_{0.8}^{-}(2) = 2$

Equal energy at every lag does not fix the Hankel spectrum or the number of dominant modes needed to explain it.
This second counterexample separates HOW MUCH energy is present from HOW that energy is organized.

05 Why this matters

TWO COMPLEMENTARY MEASURES

- Temporal depth answers the timing question: when does useful influence first become available?
- Effective dimension answers the concentration question: how many dominant Hankel modes carry that influence by horizon T?

INPUT–OUTPUT INVARIANT

- Both Dε and rτ(T) are computed from the input–output sequence $\{K_k\}$.
- They do not depend on which state basis we choose, so they are realization-invariant descriptors of memory.

CLASSIC SUMMARIES CAN MISS IT

- Matching eigenvalues, singular values, Jordan type, or lag-wise scalar energies can still miss essential memory behavior.
- Those summaries are useful, but they do not replace explicit coordinates for timing and mode concentration.

Use Dε and rτ(T) together for a fuller, more interpretable description of recurrent memory.

06 Evidence, scope & next step

Analytic
- These are exact analytic separations, not heuristic observations or trained-model artifacts.
- Result 1 shows that even if B, C, total Hankel energy, and spectral properties of A match, temporal depth can still differ substantially.
- This gives a concrete reason not to treat familiar summaries as a complete description of recurrent memory.

Checks
- The manuscript reports deterministic verification over 21 parameter settings for Result 1 and exact spectrum checks for Result 2.
- The plotted example is computed from the construction itself, so the numerical checks support the formulas rather than replace them.

Scope
- The claims here are for stable linear recurrent cores in the SISO setting.
- An important next step is to test whether analogous coordinates help interpret modern nonlinear recurrent or state-space models.

Reporting guidance

1 Report Dε to show when memory becomes available.
2 Report rτ(T) to show how concentrated that memory is across leading modes.
3 Report both together, because one can change while the other stays fixed.
4 Use eigenvalues, singular values, or scalar energies as supporting summaries, not complete substitutes.
5 This perspective makes the poster's two constructions easy to compare in one common language.

From proof to open question

The mathematical separation is established; its practical predictive value remains an open research question.

1 PROVED
Two exact counterexamples separate memory timing from Hankel mode concentration.

2 LIMITATION
Results apply to stable linear cores; trained nonlinear models are not evaluated.

3 NEXT QUESTION
Can these coordinates help explain or predict length generalization in trained models?

REFS Selected references

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[3] L. Gonon, L. Grigoryeva, and J. Ortega. "Memory and forecasting capacities of nonlinear recurrent networks." Physica D: Nonlinear Phenomena, 414:132721, 2020.